MongoDB

Database Notes — Knowledge Base

1. Overview

MongoDB is a document-oriented NoSQL database that stores data as flexible, JSON-like documents (BSON) rather than rows and tables. It is designed for high write throughput, horizontal scalability, and schema flexibility.

2. Key Concepts

- Documents — BSON objects, roughly analogous to a row, but can contain nested objects/arrays.
- Collections — groups of documents, analogous to a table but with no fixed schema.
- Sharding — built-in horizontal scaling by distributing data across multiple servers based on a shard key.
- Replica Sets — groups of servers maintaining copies of the same data for high availability and automatic failover.

3. Example Document

```
{
  "_id": "64f1...",
  "name": "Alice",
  "orders": [
     {"item": "Book", "qty": 2},
     {"item": "Pen", "qty": 5}
  ],
  "active": true
}
```

4. Advantages

- Flexible schema — good fit for evolving or heterogeneous data.
- Horizontal scalability via native sharding.
- Natural mapping to objects in application code (no ORM impedance mismatch for nested data).
- Good performance for read/write-heavy workloads on single documents.

5. Trade-offs

- Multi-document transactions exist but are more expensive than single-document operations.
- No enforced schema can lead to inconsistent data if not disciplined at the application layer.
- Complex joins across collections are less natural than in relational databases (requires $lookup aggregation).

6. When to Use

Good fit for content management systems, catalogs, real-time analytics, and applications with rapidly evolving or nested data models where relational joins are not the primary access pattern.